

**Gender Diversity, Board Independence, and CSR Committee Adoption:
The Moderating Role of Family Ownership**

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Abstract

The thesis explores the role of board governance in CSR/Sustainability Committee adoption among 845 US publicly listed firms observed between 2018 and 2022. In particular, it asks whether greater board gender diversity and board independence raise the probability of adopting a CSR/Sustainability Committee, and whether family ownership weakens these effects. Using random-effects logit models, this paper finds that firms with greater board gender diversity and a higher number of independent directors are more likely to have a CSR/Sustainability Committee, while family-controlled firms are less likely to have one. Moreover, family ownership weakens the positive effect of board gender diversity on committee adoption but leaves board independence unaffected. These findings carry implications for policymakers, investors, boards, and CSR/ESG rating agencies.

Keywords: Gender diversity, Board Independence, Family Firm, CSR Committee, Sustainability

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1. Introduction

The past two decades have seen a fundamental shift in how firms engage with the demands and expectations of society (KPMG International, 2022). Corporate Social Responsibility (CSR) has moved from being a voluntary extra activity to becoming a central part of corporate strategy. This strategic shift is driven primarily by increased stakeholder demands, regulatory requirements, and growing evidence linking CSR commitment to stronger financial performance (Cho et al., 2019; Tang et al., 2012). As a result, the majority of large firms now engage in some form of CSR activity. KPMG International (2022) therefore published a survey of sustainability reporting and found that 96% of the largest 250 companies worldwide now issue sustainability reports. This shows that reporting on social and environmental impact is now a standard practice for companies.

Many firms have gone a step further and set up specialized governance structures at the board level. The CSR/Sustainability Committee is now an important mechanism for overseeing sustainability (Danvila del Valle et al., 2019; Martinez-Ferrero et al., 2024). Its existence is consistently associated with better CSR performance: higher performance and fewer controversies in UK firms (Elmaghrabi, 2021), higher ESG (Environmental, Social & Governance) scores across Europe (Baraibar-Diez & Odriozola, 2019), and improved outcomes through CSR-linked executive pay (Radu & Smaili, 2022).

Despite this demonstrated importance, the existing literature has paid far more attention to the consequences of having a CSR committee than to the antecedents of its adoption. What drives firms to formally institutionalize CSR oversight at the board level is still a question that is largely underexplored (Martinez-Ferrero et al., 2024). This opens a meaningful gap: which governance characteristics make firms more or less likely to adopt such a committee, and further, what are the implications for governance design and CSR policy?

To examine such antecedents of a CSR/Sustainability Committee, board composition serves as a starting point. In fact, agency theory provides an interesting perspective: Boards exist to control managerial decisions on behalf of owners, and their composition determines how effectively they do so (Fama & Jensen, 1983; Jensen & Meckling, 1976). In addition to this monitoring role, boards also function as a link between the firm and important stakeholder groups such as customers, suppliers, and communities (Hillman & Dalziel, 2003). Two board characteristics are particularly relevant here: Female directors are a key factor since they bring broader stakeholder-oriented values and greater sensitivity to social risks (Boulouta, 2013;

Galbreath, 2011). These values might support more proactive sustainability agendas at the board level. Independent directors, in turn, are better positioned to support non-financial governance structures, because they are free from managerial conflicts of interest (Fama & Jensen, 1983; Islam et al., 2023). Both characteristics have been linked to stronger CSR outcomes (e.g., Agarwala et al., 2023; Bear et al., 2010; Liao et al., 2015). Evidence on whether board gender diversity (BGD) and board independence (BI) also raise the probability of adopting a CSR committee is scarce, and board independence in particular has not yet been tested as a direct predictor. Importantly, it is notable that governance mechanisms do not affect CSR in isolation or in a uniform way (Bansal et al., 2018; Guldiken et al., 2019). It therefore remains unclear whether these effects on CSR performance carry over to the decision to adopt a formal committee in US-listed firms.

One boundary condition that has received little attention is family ownership. While institutional shareholders judge a firm mainly by its financial returns, family owners also draw non-financial benefits from control, legacy, and identity, which Gómez-Mejía et al. (2007) call Socioemotional Wealth (SEW): Family firms are willing to accept worse economic outcomes to protect these endowments. Formalizing CSR governance through a dedicated committee can conflict with this desire to retain control, a tension we develop further in Section 2. This suggests that family ownership may not only reduce the likelihood of adoption, but may also weaken the positive influence that board gender diversity and board independence would otherwise exert.

This paper addresses these mentioned gaps by examining the determinants of CSR/Sustainability Committee adoption among 845 US publicly listed firms (period: 2018 to 2022). Specifically, this paper asks: Which board characteristics drive the adoption of a CSR/Sustainability Committee, and does family ownership moderate these effects? To do so, it employs random-effects logit models with a Mundlak-type test of the random-effects assumption on an unbalanced panel dataset drawn from Refinitiv Eikon.

Building on agency theory (Jensen & Meckling, 1976) and socioemotional wealth theory (Gómez-Mejía et al., 2007), this paper makes three contributions to the literature: First, research on the antecedents of CSR committee adoption is still limited (Martinez-Ferrero et al., 2024), and this study adds evidence from a US setting using recent panel data. Second, it extends prior work by introducing family ownership as a direct predictor of committee adoption. This contribution is based on SEW theory to explain why family firms are structurally less likely to

formalize CSR governance. Third, it examines whether family ownership moderates the effects of board gender diversity and board independence on committee adoption.

The remainder of this paper is organized as follows. Section 2 reviews the literature on board gender diversity, board independence, and family ownership, and develops the hypotheses. Section 3 describes the data, sample, and variables and outlines the empirical methodology. Section 4 presents the regression results. Section 5 discusses them in light of the theoretical framework and prior literature. Section 6 concludes with the main findings, the theoretical, managerial, and policy implications, the limitations, and directions for future research.

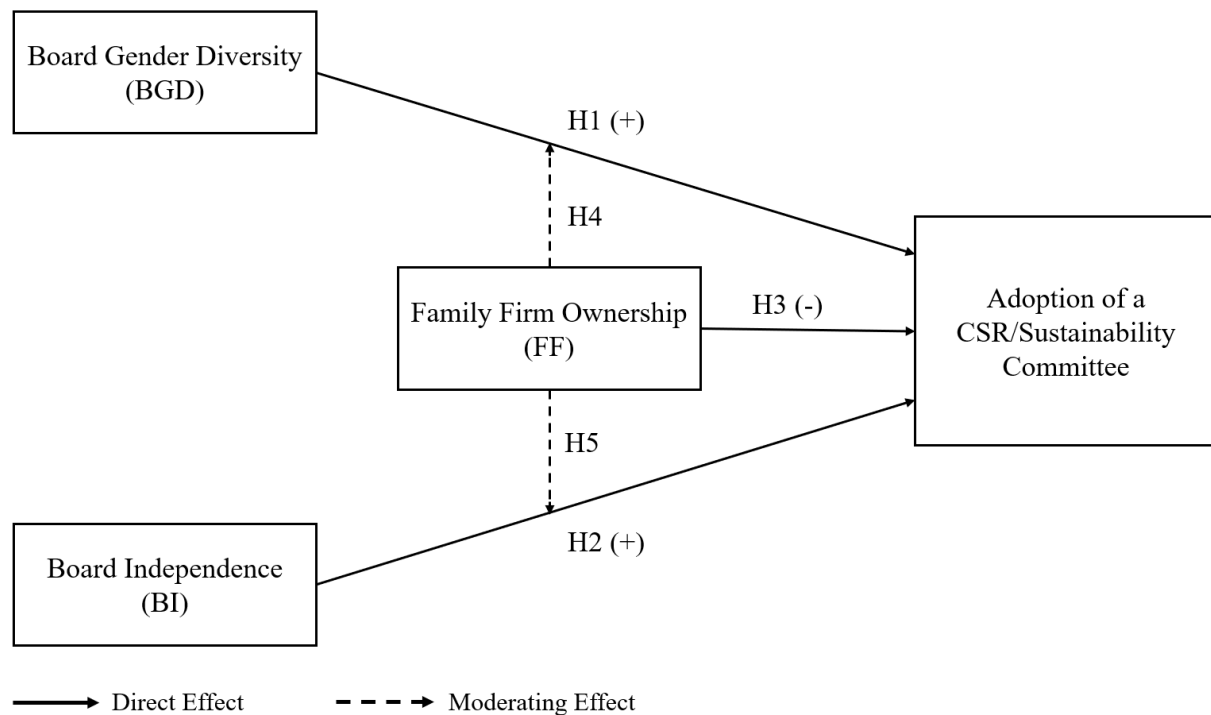
2. Literature Review

2.1 CSR and Corporate Governance

Corporate Social Responsibility (CSR) describes the responsibilities that firms have toward society, extending beyond pure profit maximization. In detail, Carroll (1991) conceptualizes CSR as a four-layered pyramid: economic, legal, ethical, and philanthropic. In practice, this means being profitable, complying with the law, acting ethically and avoiding harm, and finally being a good corporate citizen (for example by contributing resources to society). Building on this, McWilliams and Siegel (2001) define CSR more concisely as actions that appear to further some social good, beyond the firm's own interests and beyond what the law requires.

To internalize these responsibilities at the governance level, firms increasingly establish a dedicated CSR committee inside their companies. Such a committee operates at the same level as the audit or nomination committee (Danvila del Valle et al., 2019). Furthermore, it is responsible for guiding CSR policies and supporting the board in monitoring and advisory functions, especially toward shareholders and other stakeholders (Rey, 2022). Its effectiveness depends on its composition, particularly the presence of independent directors (BI) (Elmaghrabi, 2021) and board gender diversity (BGD) (Bear et al., 2010; Liao et al., 2015).

As noted, governance mechanisms do not shape CSR in isolation or uniformly (e.g., Bansal et al., 2018); their effects depend on the firm's ownership structure (Guldiken et al., 2019; Villalonga & Amit, 2006). Nevertheless, their positive effect on CSR outcomes has still been found consistently across studies (e.g., Baraibar-Diez & Odriozola, 2019; Elmaghrabi, 2021). Drawing on agency theory (Jensen & Meckling, 1976) and socioemotional wealth theory (Gómez-Mejía et al., 2007), this study develops and tests five hypotheses (Figure 1):

Figure 1: Conceptual Research Model

2.2 Hypotheses

2.2.1 Gender-Diverse Boards as Drivers of CSR/Sustainability Committee Adoption

Gender represents one of the key dimensions of board diversity (Rao & Tilt, 2016), and its influence on corporate sustainability outcomes has been extensively documented in the literature: Female directors tend to bring broader stakeholder-oriented values and greater sensitivity to social risks (Boulouta, 2013; Galbreath, 2011). From an agency theory perspective, the separation of ownership and control creates agency costs (Jensen & Meckling, 1976), and boards exist to limit these costs by ratifying and monitoring managerial decisions on behalf of owners (Fama & Jensen, 1983). The board's composition shapes how effectively this monitoring function is exercised: Adams and Ferreira (2009) find that gender-diverse boards engage more actively in monitoring. Specifically, such engagement is reflected in better director attendance and a greater likelihood of female directors joining monitoring committees. In sum, these value and monitoring capacities suggest that gender-diverse boards are structurally better positioned to champion formal governance structures oriented toward stakeholder accountability.

In addition, this monitoring capacity results in measurable CSR performance outcomes across multiple studies: Bear et al. (2010) find that a higher proportion of female directors significantly

improves CSR strength ratings among 51 large US firms. Similarly, Deschênes et al. (2015) document a positive association between women's board representation and CSR scores among the 60 largest Canadian firms. Moreover, Liao et al. (2015) show that gender diversity is positively associated with greenhouse gas disclosure. An important nuance is added by Elmaghrabi (2021): The study finds, by analyzing FTSE 100 non-financial firms, that a female CSR committee chair is positively and significantly associated with CSR performance, whereas the mere proportion of female committee members is not. This suggests that it is not simply the presence of women on the board but their active involvement in sustainability governance structures that drives meaningful outcomes.

Recent research has moved from CSR performance to the governance structures behind it, asking whether gender-diverse boards are more likely to formalize CSR by setting up a dedicated CSR/Sustainability Committee. Martinez-Ferrero et al. (2024) offer the most direct evidence here: They find that board gender diversity significantly increases the probability that a firm establishes a CSR committee. Importantly, these results can be explained by critical mass theory: Female directors can exert meaningful collective influence on governance decisions only once their number on the board reaches a critical mass, commonly set at three or more women (Kanter, 1977). Above this threshold, they can drive structural changes such as establishing a dedicated sustainability committee (Martinez-Ferrero et al., 2024) and improve substantive CSR outcomes (Post et al., 2011).

However, this relationship is not as straightforward as the accumulated evidence on CSR performance might suggest. Several studies introduce important warning findings. Guldiken et al. (2019), for instance, show that many boards stop at a single female director, a pattern they describe as potential tokenism rather than a real diversification of board composition. They find that additional female appointments become more likely when female top managers are present and when the sole female director serves on the nominating committee. Where these conditions are absent, female directors may lack the critical mass, positional authority, or organizational support needed to drive structural changes in formal CSR governance. In addition, Adams and Ferreira (2009) note that this effect depends on context. Gender diversity improves governance outcomes mainly in firms with weak shareholder rights, where additional monitoring adds value, and may be less effective where governance is already strong.

Taken together, the overall evidence still points to a positive baseline effect. The concerns raised (Adams & Ferreira, 2009; Guldiken et al., 2019) are better understood as factors that may weaken the relationship in specific contexts, not reverse it. We therefore propose:

***H1:** Firms with greater board gender diversity are more likely to have a CSR/Sustainability Committee.*

2.2.2 Independent Boards as Catalysts of CSR/Sustainability Committee Adoption

As outlined in the introduction, the board controls managerial decisions in the interest of the firm's owners (Jensen & Meckling, 1976). In addition, Fama and Jensen (1983) further distinguish decision management by insiders from decision control, the board's ratification and monitoring of those decisions. Independent directors are better suited to control decisions because they are not influenced by conflicting ties with management. Beyond monitoring, they contribute an external perspective and press for decisions that account for broader stakeholder interests (Islam et al., 2023).

Board independence is also linked to stronger CSR outcomes. Agarwala et al. (2023) find that greater board independence is associated with higher CSR performance, and Meqbel et al. (2025), using a global sample across 44 countries, show that CSR committee independence is positively associated with environmental innovation. Independent members reduce managerial opportunism and bring in external expertise, which strengthens the committee's capacity to act on environmental priorities. Independence also matters for committee leadership: Elmaghrabi (2021) finds that CSR committees with a majority of independent members perform better, and that more independent directors combined with a CSR-expert chair are associated with fewer controversies. Moreover, Jarboui et al. (2023) show that independent directors on CSR committees drive stronger CSR performance overall.

Taken together, these findings point in the same direction. Board independence strengthens a firm's commitment to CSR across multiple dimensions (Elmaghrabi, 2021; Jarboui et al., 2023; Meqbel et al., 2025). Whether it also leads to a formal CSR governance structure such as a CSR/Sustainability Committee remains largely underexplored. Given the theory and evidence reviewed above, we expect the relationship to extend to committee formation. We therefore propose:

***H2:** Firms with greater board independence are more likely to have a CSR/Sustainability Committee.*

2.2.3 Family Control as a Barrier to CSR/Sustainability Committee Adoption

Family firms are fundamentally different from non-family firms, not just in their ownership structure but also in how they make decisions (Chua et al., 1999). Key differences from non-family firms emerge when comparing financial and non-financial goals. In particular, family owners tend to prefer several non-financial benefits from their firms: the ability to exercise control, the continuation of a family legacy, and the preservation of the firm's identity as a family enterprise. Importantly, a key nuance of this framework, known as socioemotional wealth (SEW), states that family firms may accept worse economic outcomes in order to preserve these non-financial benefits (Gómez-Mejía et al., 2007).

These characteristics can directly influence how family firms approach formal governance structures. Characteristics of a CSR committee include external oversight, transparency, and monitoring of sustainability and governance practices (Rey, 2022). Family firms are therefore not necessarily less committed to CSR but tend to prefer managing it informally rather than through formal structures that limit their control: Cruz et al. (2014) find in a sample of 598 listed European firms that family ownership is negatively associated with governance-related CSR scores. However, no significant difference emerges for external dimensions such as environmental performance. This confirms that family firms do not avoid CSR as such but resist the formal structures that limit their control.

The empirical evidence on family firms and CSR outcomes is, however, far from settled. Several studies report stronger social performance in family firms: Dyer and Whetten (2006) find that family firms in the S&P 500 are significantly better at avoiding social concerns, especially regarding employees, the environment, and product safety. This is mainly because families cannot distance themselves from their firm's reputation the way professional managers can. Furthermore, Campopiano and De Massis (2015) add that family firms tend to publish a wider variety of CSR reports. In particular, their reports focus on environmental and philanthropic themes, mirroring their desire for legitimacy within their communities. Aoi et al. (2015) similarly show that the presence of a family CEO can positively affect corporate social performance, though this effect is industry-contingent, holding in light manufacturing (foods, textiles, pharmaceuticals) and service sectors (retail, wholesale). However, in heavy industries the effect is shown to be negative, which reinforces the context dependence of such governance structures.

The majority of research, however, finds that higher family ownership is associated with lower CSR performance (Dal Maso et al., 2020; Labelle et al., 2018; Latella et al., 2026;

Miroshnychenko et al., 2022). In addition, Adardour et al. (2026) find that the risk-reducing effect of ESG performance is significantly weaker in family firms than in non-family firms, suggesting that family firms do not effectively leverage formal ESG governance mechanisms to manage financial risk.

Regarding the role of the CSR committee, it serves as a central mechanism for channeling board resources and expertise into structured sustainability practices (Meqbel et al., 2025), a mechanism family firms tend to avoid. Moisello et al. (2026) therefore find that family ownership and family presence on the board are negatively correlated with ESG disclosure, confirming this idea.

While the relationship between family ownership and CSR activities has been well studied, the studies reviewed above focus on CSR performance or ESG disclosure rather than on the existence of a CSR/Sustainability Committee. Whether family firms are more or less likely to have such a committee therefore remains largely unexplored (Martinez-Ferrero et al., 2024). Addressing this research gap, we therefore propose:

H3: *Family firms are less likely to have a CSR/Sustainability Committee than non-family firms.*

2.2.4 Family Ownership as a Boundary Condition for Gender-Driven CSR/Sustainability Committee Adoption

While the prevailing evidence consistently supports a positive relationship between board gender diversity (BGD) and the adoption of a CSR/Sustainability Committee, the literature also reveals important boundary conditions that qualify this relationship: Martinez-Ferrero et al. (2024) find that board gender diversity increases the likelihood of CSR committee adoption, but that this effect weakens when female directors are also independent. This suggests that gender diversity and board independence do not simply reinforce each other. Furthermore, Grau Grau et al. (2025) show that country-level gender parity negatively moderates the BGD–ESG relationship. Also, Sánchez-Hernández et al. (2025) demonstrate that the mere presence of female directors is insufficient. Rather, it is only their active involvement within a sustainability committee that yields meaningful effects on reporting quality. Summarizing, these findings raise a fundamental question: Under what organizational conditions does the positive influence of gender diversity on formal CSR governance structures become weaker or even reversed?

We argue that family ownership represents one such boundary condition. Formalizing CSR governance can directly conflict with family control, as research shows: Dick et al. (2021) show that founder-controlled family firms exhibit lower CSR engagement, reflecting the family's motivation to protect control, while Terlaak et al. (2018) argue that family owners weigh the reputational benefits of environmental disclosure against the risk of losing control. This suggests that family ownership discourages the formal establishment of a CSR/Sustainability Committee, since it demands stronger governance commitment than families typically prefer. Family owners therefore have both the willingness and the ability to refuse formal CSR governance structures, regardless of how the board is composed (Chrisman et al., 2012; De Massis et al., 2014). Willingness arises from the family's interest in preserving socioemotional wealth (Berrone et al., 2012; Gómez-Mejía et al., 2007). This matters most for board gender diversity, because its positive influence on CSR governance works through monitoring intensity and stakeholder voice (Adams & Ferreira, 2009), and these are the channels family control can neutralize. Ability refers to the family's power to shape governance outcomes in practice (De Massis et al., 2014). In firms with concentrated ownership, families control who is appointed to the board in the first place. This allows them to limit the influence of outside directors, including female directors, from the outset (Anderson & Reeb, 2004).

Interestingly, Cordeiro et al. (2020) find the opposite: Family owners and dual-class owners (e.g., owners holding shares with superior voting rights) appoint female directors who share their views, which strengthens rather than weakens their effect on environmental performance. This suggests that family owners support female directors when it serves their own interests but resist their influence on structural changes such as a CSR/Sustainability Committee. Oh et al. (2019) report a similar pattern: In family-managed firms, boards tend to serve as impression management tools rather than exert real influence on monitoring.

The evidence on this moderating relationship is therefore not fully settled. For example, Houcine and Derouiche (2024) find in a sample of French listed firms that family ownership positively moderates the effect of board gender diversity on corporate social performance, suggesting that female directors may retain meaningful influence even under family control. Importantly, this study captures CSR performance outcomes rather than the adoption of formal governance structures. A CSR committee, however, reduces the family's control over decisions. SEW theory predicts that family owners resist such structural changes (Berrone et al., 2012; Gómez-Mejía et al., 2007), even if they still support CSR informally (Cruz et al., 2014). Based on these prior findings and theories, we expect a negative influence of family ownership on the

relationship between gender diversity and the establishment of a CSR/Sustainability Committee. Therefore, we propose:

H4: *Family firm ownership weakens the positive effect of board gender diversity on the likelihood of having a CSR/Sustainability Committee.*

2.2.5 Family Ownership as a Boundary Condition for Independence-Driven CSR/Sustainability Committee Adoption

Similarly, just as the positive effect of board gender diversity on CSR committee adoption depends on governance context, the same logic holds for board independence. Independent directors are generally expected to broaden the firm's stakeholder orientation and strengthen external accountability (Hillman & Dalziel, 2003; Islam et al., 2023). Prior research links board independence to stronger CSR engagement (e.g., Agarwala et al., 2023; Meqbel et al., 2025). Specifically, this also holds for the independence of dedicated sustainability committees (Jarboui et al., 2023). However, whether and to what extent independent directors are able to convert this orientation into formal governance structures depends largely on the ownership context in which they operate (Backman & Palmberg, 2015; Bansal et al., 2018).

Under family ownership, this environment shifts considerably: As outlined in Section 2.2.3, family owners' SEW priorities outweigh purely financial or governance considerations (Gómez-Mejía et al., 2007). A committee limits the control they want to protect (Berrone et al., 2012), so they are likely to resist it even when independent directors support its formation. Anderson and Reeb (2004) show how: Through the nominating committee, families keep independent directors to only 43.9% of board seats in S&P 500 family firms, substantially fewer than in non-family firms. Beyond limiting the number of independent directors, families also shape who those directors are: Villalonga and Amit (2006) show that family firms have significantly fewer independent directors, and that their presence does not significantly affect firm value.

These constraints on the nominating committee have consequences: Oh et al. (2019) find that the positive effect of outside directors on CSR outcomes holds when family involvement is weak but reverses when it is strong. Anwar and Ahmed (2020) report the same pattern in the Chinese market: Board independence is positively associated with CSR performance in non-family firms, and the effect is significantly weaker under family control. Both findings point in

the same way: The family's governance logic overrides the monitoring role of independent directors.

In sum, these findings clearly show that the relation between Board Independence and CSR practices/outcomes varies between family and non-family firms. Nevertheless, this evidence concerns CSR performance and disclosure rather than formal governance structures. Whether family ownership also moderates the effect of board independence on the existence of a CSR/Sustainability Committee remains, to the best of our knowledge, untested. Building on this, we propose:

H5: *Family firm ownership weakens the positive effect of board independence on the likelihood of having a CSR/Sustainability Committee.*

3. Methodology

3.1 Research Design and Data

This study employs a quantitative, deductive, and observational research design testing theory-derived hypotheses using secondary panel data on publicly listed firms. A longitudinal panel structure covering 2018 to 2022 is used, as it captures both variation across firms and changes within the same firm over time. The unit of analysis is the firm-year observation.

The empirical analysis is carried out on firm-level data obtained from Refinitiv Eikon, a database providing ESG metrics, board-level governance indicators, and financial information for publicly listed companies. The initial dataset comprises 998 US publicly listed firms observed across all five years of the study period, yielding a balanced panel of 4,990 firm-year observations before cleaning. The sample varies over 18 industries based on the North American Industry Classification System (NAICS). Manufacturing represents the largest share (33.8%), followed by Finance and Insurance (14.5%), Retail Trade (8.3%), and Information (7.6%) (see Appendix B). Financial variables such as total assets, leverage, and firm age were sourced from the same database.

Refinitiv Eikon does not provide a standardized variable for family ownership, so the classification was coded manually. For each firm, the DEF 14A proxy statements filed with the Securities and Exchange Commission (SEC) were reviewed against two criteria: A firm is classified as family-owned if a founding family or family group holds at least 5% of the outstanding shares and at least one family member serves on the board of directors or in a

management position, following established definitions in the family business literature (Chua et al., 1999; Villalonga & Amit, 2006). Manual coding identifies family firms more accurately than automated or threshold-based proxies, which are prone to misclassification (Amit & Villalonga, 2014). The variable equals 1 for family firms and 0 otherwise.

Firms with fewer than three usable observations after the one-year lag were then excluded, leaving an unbalanced intermediate panel of 876 firms. To account for missing values, listwise deletion was applied to the remaining explanatory and control variables, which reduced the panel to 846 firms and 2,841 firm-year observations. One industry (Educational Services) was then dropped because it predicted the outcome perfectly, leaving a final estimation sample of 845 firms and 2,837 firm-year observations (or 3.4 observations per firm on average). A firm can contribute at most four observations, because the one-year lag restricts the dependent variable to the years 2019 to 2022 (see Appendix A).

3.2 Variable Description

3.2.1 Dependent Variable

CSR/Sustainability Committee. The dependent variable captures whether a specialized CSR/Sustainability Committee exists in a given firm-year. Such committees are responsible for overseeing sustainability and corporate social responsibility. The binary variable takes a value of 1 if such a committee exists within the firm's governance structure and 0 otherwise. The variable is sourced from Refinitiv Eikon's ESG database, which collects this information annually from firm disclosures and sustainability reports. Binary indicators like this are standard in CSR governance research (Baraibar-Diez & Odriozola, 2019; Martinez-Ferrero et al., 2024; Radu & Smaili, 2022).

3.2.2 Independent Variables

Three governance variables serve as explanatory variables in this study: Board Gender Diversity, Board Independence, and Family Firm.

Board Gender Diversity. Board gender diversity is measured by the percentage of female directors on the board, as reported in Refinitiv Eikon. A higher value reflects a greater proportion of female members on the board, consistent with prior research linking board

diversity to CSR engagement (Bear et al., 2010). To address reverse causality, the variable is lagged by one period (t-1).

Board Independence. Board independence is measured by the percentage of independent directors on the board, drawn from Refinitiv Eikon. An independent director has no business ties to the firm beyond the board seat itself (Anderson & Reeb, 2004). Moreover, such independent directors are associated with stronger monitoring functions and a greater emphasis on stakeholder-oriented governance (Hillman & Dalziel, 2003). The variable is lagged by one period (t-1).

Family Firm. As defined in Section 3.1, a firm is classified as family-owned if a founding family or family group holds an ownership stake of at least 5% of the firm's outstanding shares and at least one family member serves on the board of directors or in a management position. This threshold also reflects the minimum ownership level above which beneficial owners must be disclosed in a firm's proxy statement (17 C.F.R. § 229.403, 2026). The binary variable is lagged by one period (t-1).

Beyond its role as a direct predictor, the family firm variable also serves as a moderator in two interaction specifications, capturing whether the family governance structure moderates the positive effects of board gender diversity and board independence on the adoption of a CSR/Sustainability Committee.

To examine this moderation, two interaction terms are constructed. Both continuous governance variables, Board Gender Diversity and Board Independence, are first centered around their mean. Then this term is multiplied by the family firm indicator. All three variables enter at t-1, so the interaction terms are lagged by one period (t-1) as well.

3.2.3 Control Variables

Firm Size. Firm size is measured as the natural logarithm of total assets, sourced from Refinitiv Eikon. The logarithmic transformation reduces the skewness that comes with major differences in firm size. Larger firms tend to face greater stakeholder scrutiny and have more resources available to establish formal CSR governance structures (Cowen et al., 1987; Rao & Tilt, 2016). The variable is lagged by one period (t-1).

Leverage. Leverage is measured as the ratio of net debt to EBITDA (Earnings Before Interest, Taxes, Depreciation and Amortization). It reflects the firm's indebtedness relative to its

operating earnings and is a common proxy for firm risk, a standard control in research on corporate social performance (Boulouta, 2013). Higher leverage may signal financial constraints that limit investment in voluntary governance structures. The variable is lagged by one period (t-1).

Firm Age. Firm age is measured as the number of years since the firm's founding, as recorded in Refinitiv Eikon. Older firms may have more entrenched governance routines and established stakeholder relationships (Cochran & Wood, 1984; Oh et al., 2019; Villalonga & Amit, 2006). Such structures can affect the likelihood of formalizing CSR oversight through a dedicated committee. The variable is lagged by one period (t-1).

Industry. Industry is controlled for using binary indicator variables based on the North American Industry Classification System (NAICS). Prior research shows that industry membership is systematically associated with differences in CSR practices and reporting (Aoi et al., 2015; Rao & Tilt, 2016). Including industry fixed effects ensures that estimated coefficients reflect within-industry variation rather than cross-industry differences in CSR governance practices.

Year. Year fixed effects are included as binary indicator variables for each year in the sample period. They control for common time trends that affect all firms at once, such as shifts in regulatory requirements, investor expectations, or sustainability norms. This should therefore prevent temporal trends in CSR governance from influencing the results.

3.3 Descriptive Analysis

Table 1 reports the descriptive statistics of the main variables included in the analysis. The dependent variable, CSR Committee (csrcom), is a binary indicator equal to 1 if a dedicated sustainability committee exists at the board level and 0 otherwise. The mean of 0.608 indicates that approximately 61% of the firm-year observations in the sample have established such a committee.

Regarding the independent variables, in the case of the Board Gender Diversity, the mean of 25.33% indicates that, on average, about one in four board seats is held by female directors. On the other hand, Board Independence has a mean of 81.88%, which shows that the majority of board members in the sample are classified as independent directors. The third main

independent variable, Family Firm, has a mean of 0.189, indicating that approximately 19% of firm-year observations in the sample are classified as family-controlled firms.

The control variables reveal the following observations: Firm Size (measured as the natural logarithm of total assets) has a mean of 23.16. Leverage (Net Debt/EBITDA) has a mean of 3.94, and values ranging from -108.57 to 1,773.56, suggesting substantial heterogeneity in capital structures across the sample. Next, Firm Age has a mean of 67.12 years, ranging from newly founded firms to companies with over two centuries of existence.

Table 1. Descriptive Statistics

Variable	N	Mean	SD	Min	Max
CSR Committee (csrcom)	3,901	0.608	0.488	0	1
Board Gender Diversity (genderdiv, %)	3,904	25.328	10.467	0	75
Board Independence (indbm, %)	3,904	81.876	12.055	0	100
Family Firm (famili)	4,302	0.189	0.392	0	1
Firm Size (ln_ta, log total assets)	4,317	23.157	1.470	10.13	28.95
Leverage (netdebtebitda, Net Debt/EBITDA)	4,198	3.943	34.751	-108.57	1,773.56
Firm Age (age, years)	4,380	67.117	47.403	0	230

Note: All variables measured at firm-year level. Observations vary due to missing values (see Section 3.1).

Table 2 presents two-sample t-tests comparing family and non-family firms across all key variables of the analysis. The results reveal statistically significant differences across all dimensions: In the case of the dependent variable, non-family firms show a higher likelihood of having a CSR/Sustainability Committee, with a mean of 0.654 compared to 0.425 for family firms. This difference provides preliminary support for H3. For the governance variables, non-family firms exhibit higher Board Gender Diversity (26.12% vs. 22.04%) and substantially higher Board Independence (84.19% vs. 73.00%). As for the control variables, non-family firms are significantly larger in terms of total assets (23.22 vs. 22.87 log points) and older on average (70.60 vs. 54.11 years). Family firms, on the other hand, show higher average leverage (Net

Debt/EBITDA = 6.27 vs. 3.39), indicating greater financial constraints relative to their non-family counterparts.

Table 2. Comparison of Family and Non-Family Firms

Variable	Non-Family (famili = 0)		Family (famili = 1)		t-stat	p
	N	Mean	N	Mean		
CSR Committee (csrcom)	3,092	0.654	744	0.425	11.72	0.000
Board Gender Diversity (%)	3,095	26.115	744	22.044	9.66	0.000
Board Independence (%)	3,095	84.188	744	73.001	24.74	0.000
Firm Size (ln Total Assets)	3,443	23.224	801	22.873	6.15	0.000
Leverage (Net Debt/EBITDA)	3,369	3.389	766	6.266	-2.06	0.040
Firm Age (years)	3,488	70.604	814	54.106	9.01	0.000

Note: Two-sample t-test with equal variances. p = two-tailed p-value.

The pairwise Pearson correlations (Table 3) reveals the following observations: Among the most notable findings, CSR Committee shows its strongest positive correlation with Firm Size ($r = 0.348$), followed by Board Gender Diversity ($r = 0.277$) and Board Independence ($r = 0.260$). This observation shows that larger and better-governed firms are more likely to have a CSR/Sustainability Committee. Furthermore, Family Firm is negatively correlated with CSR Committee ($r = -0.186$). This value reinforces the group differences reported in Table 2. The strongest correlation among the independent variables is between Family Firm and Board Independence ($r = -0.371$). This shows that family firms tend to have less independent boards. No correlation among the independent variables exceeds 0.40 in absolute value, suggesting no evident multicollinearity problems in the main models.

Table 3. Pearson Correlation Matrix

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(1) CSR Committee	1.000						
(2) Gender Diversity	0.277*	1.000					
(3) Board Independence	0.260*	0.275*	1.000				
(4) Family Firm	-0.186*	-0.154*	-0.371*	1.000			
(5) Firm Size	0.348*	0.183*	0.170*	-0.094*	1.000		
(6) Leverage	-0.028	0.027	-0.012	0.032*	0.033*	1.000	
(7) Firm Age	0.145*	0.111*	0.145*	-0.136*	0.201*	-0.005	1.000

Note: * $p < 0.05$ (two-tailed). Pearson correlations. N varies due to missing values.

3.4 Estimation Method

To examine the regression models, this study employs random-effects logit models. The dependent variable is a binary outcome equal to 1 if a firm has an established CSR/Sustainability Committee in a given year and 0 otherwise.

Random-effects estimation assumes that the unobserved firm effect is uncorrelated with the explanatory variables (Greene, 2020; Wooldridge, 2010). If this assumption is violated, the coefficients are biased, because part of the effect of the unobserved firm characteristic is attributed to the observed variables. To assess whether this assumption holds, the specification is augmented following the Mundlak approach: The within-firm means of all time-varying explanatory variables are added as additional controls. These means capture the part of the unobserved firm effect that is correlated with the explanatory variables, which reduces the remaining bias (Mundlak, 1978). Testing whether these means are jointly significant also serves as a Hausman-type test of the random-effects assumption. This test is reported in Section 4.1; the main models are estimated without these means.

The choice of random-effects estimation is motivated by the limited within-firm variation of the key explanatory variables. For example, family ownership exhibits a within standard deviation of only 0.035 compared to a between standard deviation of 0.393, indicating that

family status remains largely stable over time (see Appendix D). Board independence and gender diversity similarly display substantially greater between-firm than within-firm variation. Because the hypotheses in this study focus on governance characteristics that are highly persistent over time, a fixed-effects specification would absorb much of the relevant information and substantially reduce estimation efficiency. Moreover, the conditional fixed-effects logit estimator discards all firms in which the dependent variable shows no within-firm variation, which in this sample would result in a substantial loss of observations. Random-effects logit models are therefore deemed more appropriate. This is consistent with standard econometric guidance: Random-effects estimators allow time-invariant or highly persistent regressors to be included, which fixed-effects models do not (Greene, 2020; Wooldridge, 2010). In the corporate governance literature, similar specifications are employed when key variables such as ownership structure or family control exhibit low temporal variation (Anderson & Reeb, 2003; Villalonga & Amit, 2006).

To reduce reverse causality concerns, all governance variables and firm-level controls are lagged by one year ($t-1$), ensuring that the regressors precede the dependent variable in time. The continuous governance variables are mean-centered before constructing the interaction terms with the family firm indicator, which reduces multicollinearity. All models include year and industry fixed effects, specified as binary indicators (dummy variables), to account for common time trends and sector-specific heterogeneity. Building on this specification, the following four models were developed:

M1 tests the direct effects of the governance variables without interaction terms (H1–H3):

$$\begin{aligned} \text{logit}[P(\text{CSR}COM_{it} = 1)] = & \beta_0 + \beta_1 \text{GenderDiversity}_{i,t-1} + \beta_2 \text{BoardIndependence}_{i,t-1} \\ & + \beta_3 \text{FamilyFirm}_{i,t-1} + \beta_4 \text{FirmSize}_{i,t-1} + \beta_5 \text{Leverage}_{i,t-1} \\ & + \beta_6 \text{FirmAge}_{i,t-1} + \text{Year}_t + \text{Industry}_i + u_i + \varepsilon_{it} \end{aligned}$$

M2 adds the Family Firm $\times$ Gender Diversity interaction to test H4:

$$\begin{aligned} \text{logit}[P(\text{CSR}COM_{it} = 1)] = & \beta_0 + \beta_1 \text{GenderDiversity}(c)_{i,t-1} + \beta_2 \text{BoardIndependence}(c)_{i,t-1} \\ & + \beta_3 \text{FamilyFirm}_{i,t-1} + \beta_4 (\text{FamilyFirm} \times \text{GenderDiversity}(c))_{i,t-1} \\ & + \beta_5 \text{FirmSize}_{i,t-1} + \beta_6 \text{Leverage}_{i,t-1} + \beta_7 \text{FirmAge}_{i,t-1} \\ & + \text{Year}_t + \text{Industry}_i + u_i + \varepsilon_{it} \end{aligned}$$

M3 adds the Family Firm $\times$ Board Independence interaction to test H5:

$$\begin{aligned} \text{logit}[P(\text{CSR}COM_{it} = 1)] = & \beta_0 + \beta_1 \text{GenderDiversity}(c)_{i,t-1} + \beta_2 \text{BoardIndependence}(c)_{i,t-1} \\ & + \beta_3 \text{FamilyFirm}_{i,t-1} + \beta_4 (\text{FamilyFirm} \times \text{BoardIndependence}(c))_{i,t-1} \\ & + \beta_5 \text{FirmSize}_{i,t-1} + \beta_6 \text{Leverage}_{i,t-1} + \beta_7 \text{FirmAge}_{i,t-1} \\ & + \text{Year}_t + \text{Industry}_i + u_i + \varepsilon_{it} \end{aligned}$$

M4 combines both interactions to test H4 and H5 jointly:

$$\begin{aligned} \text{logit}[P(\text{CSR}COM_{it} = 1)] = & \beta_0 + \beta_1 \text{GenderDiversity}(c)_{i,t-1} + \beta_2 \text{BoardIndependence}(c)_{i,t-1} \\ & + \beta_3 \text{FamilyFirm}_{i,t-1} + \beta_4 (\text{FamilyFirm} \times \text{GenderDiversity}(c))_{i,t-1} \\ & + \beta_5 (\text{FamilyFirm} \times \text{BoardIndependence}(c))_{i,t-1} + \beta_6 \text{FirmSize}_{i,t-1} \\ & + \beta_7 \text{Leverage}_{i,t-1} + \beta_8 \text{FirmAge}_{i,t-1} \\ & + \text{Year}_t + \text{Industry}_i + u_i + \varepsilon_{it} \end{aligned}$$

4. Empirical Results

4.1 Hausman Test (Mundlak Approach)

The null hypothesis of the Hausman test (Mundlak approach) is that the coefficients of the firm-specific means are jointly zero, which would indicate that the unobserved firm effect is uncorrelated with the explanatory variables and that the random-effects assumption holds. The results reject this hypothesis ($\chi^2(6) = 50.64$, $p < 0.001$), suggesting that unobserved firm characteristics are related to the explanatory variables (see Appendix E). As argued above, despite this finding, the random-effects specification is retained. This is because a fixed-effects model would rely exclusively on within-firm variation, which is limited for key variables such as family firm status ($\text{within-SD} = 0.035$), and would therefore discard most of the relevant information.

4.2 Direct Effects

Table 4 presents the empirical results for Hypotheses 1, 2, and 3. Model 1 tests the direct effects of all governance variables on the probability of a firm having a CSR/Sustainability Committee.

First, Hypothesis 1 states that a higher level of board gender diversity increases the probability of a firm having a CSR/Sustainability Committee. The coefficient of Board Gender Diversity

is positive and significant at the 1% level ($b = 0.100, p < 0.001$). The results confirm that increased gender diversity on the board is associated with a higher probability of adopting a CSR/Sustainability Committee. Therefore, H1 is supported.

Similarly, Hypothesis 2 states that a higher level of board independence increases the probability of a firm having a CSR/Sustainability Committee. The coefficient of Board Independence is positive and significant at the 1% level ($b = 0.042, p < 0.001$). The results show that firms with a higher rate of independent directors are more likely to have a CSR/Sustainability committee in place, supporting H2.

In contrast, Hypothesis 3 states that family firms are less likely to have a CSR/Sustainability Committee than non-family firms. The coefficient of Family Firm is negative and significant at the 1% level ($b = -2.654, p < 0.001$). This implies that family-controlled firms have substantially lower odds of having a CSR/Sustainability Committee relative to their non-family counterparts, supporting H3.

Regarding the control variables, Firm Size appears as an important predictor of CSR/Sustainability Committee adoption with a strong positive effect ($b = 2.534, p < 0.001$). This result is consistent with the expectation that larger firms face greater public and stakeholder scrutiny (Cowen et al., 1987) and typically have greater access to resources to formalize CSR governance (Udayasankar, 2008). Moreover, Firm Age also appears to be positively and significantly associated with CSR/Sustainability Committee adoption ($b = 0.009, p < 0.01$). However, the coefficient of Leverage is not significant ($p = 0.796$) (see Appendix F).

Table 4. RE Logit Results: CSR Committee (Log-Odds Coefficients)

Variable	M1	M2	M3	M4
	Direct	H4	H5	Full
Board Gender Diversity, t-1	0.100*** (0.013)	0.118*** (0.015)	0.100*** (0.013)	0.120*** (0.017)
Board Independence, t-1	0.042***	0.044***	0.039***	0.037**

Variable	M1	M2	M3	M4
	Direct	H4	H5	Full
	(0.011)	(0.011)	(0.013)	(0.015)
Family Firm	-2.654***	-2.887***	-2.580***	-2.853***
	(0.343)	(0.345)	(0.365)	(0.426)
Family x Gender Diversity, t-1		-0.080***		-0.092***
		(0.028)		(0.033)
Family x Board Independence, t-1			0.014	0.033
			(0.023)	(0.028)
Firm Size (ln Total Assets), t-1	2.534***	2.560***	2.539***	2.586***
	(0.125)	(0.125)	(0.125)	(0.148)
Leverage (Net Debt/EBITDA), t-1	-0.001	-0.000	-0.001	-0.000
	(0.002)	(0.002)	(0.002)	(0.002)
Firm Age (years), t-1	0.009***	0.008***	0.008***	0.009**
	(0.003)	(0.003)	(0.003)	(0.004)
N	2,837	2,837	2,837	2,837

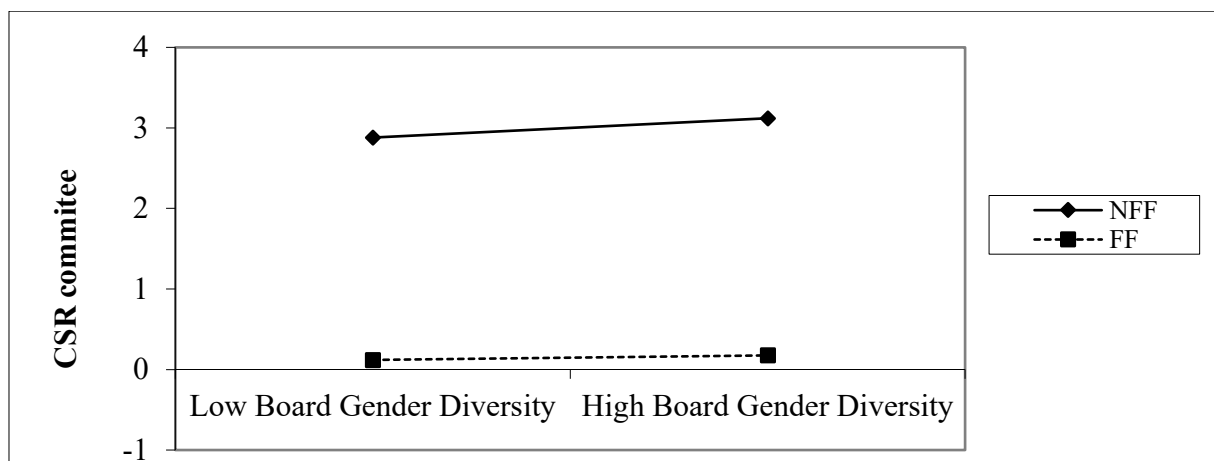
*Random-effects logit (xtlogit, re, intpoints(7)). Log-odds coefficients; standard errors in parentheses. Industry and year fixed effects included in all models. M2-M4 use mean-centered governance variables for interaction terms. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

4.3 Moderation Effects

Table 4 further presents the results for Hypotheses 4 and 5, examining whether family firm ownership moderates the positive effects of board gender diversity and board independence on CSR/Sustainability Committee adoption.

First, Hypothesis 4 states that family firm ownership weakens the positive effect of board gender diversity on the likelihood of having a CSR/Sustainability Committee. The coefficient of the interaction term “Family x Gender Diversity” appears negative and significant across multiple models/settings: In the isolated model (Model 2), the interaction term is negative and significant at the 1% level ($b = -0.080, p < 0.01$). Furthermore, the interaction term shows related results in the full model (Model 4), ($b = -0.092, p < 0.01$). These results imply that the positive effect of board gender diversity on CSR/Sustainability Committee adoption is significantly weaker in family-controlled firms. H4 is therefore supported.

Figure 2: Moderating Effect of Family Ownership



H4: Moderating effect of family ownership on the relationship between board gender diversity and CSR committee adoption. (Values used from M4)

Figure 2 illustrates this result further. For non-family firms (NFF), the slope is clearly positive: Higher board gender diversity is associated with a substantially greater likelihood of having a CSR/Sustainability Committee. In contrast, for family firms (FF), the slope is nearly flat, suggesting that female board representation has little bearing on whether a CSR/Sustainability Committee is established. The divergence between the two slopes becomes more pronounced at higher levels of gender diversity. Figure 2 therefore visually confirms that family ownership weakens the positive effect of board gender diversity on CSR/Sustainability Committee adoption.

In contrast to Hypotheses 1 to 4, Hypothesis 5 yields different results: It states that family firm ownership weakens the positive effect of board independence on the likelihood of having a

CSR/Sustainability Committee. In the isolated model (Model 3), the coefficient of the interaction term "Family x Board Independence" is not significant ($b = 0.014$, $p = 0.537$). Moreover, testing the effect in the full model (Model 4) confirms this result: The coefficient is positive, but also not significant ($b = 0.033$, $p = 0.234$). These findings show that there is no statistically significant moderation effect of family ownership on the relationship between board independence and the existence of a CSR/Sustainability Committee. H5 is therefore not supported.

4.4 Summary of Hypothesis Tests

Table 5 provides an overview of the hypothesis test results. Of the five hypotheses proposed, four are supported by empirical evidence. The direct effects of board gender diversity, board independence, and family firm ownership on the probability of a firm having a CSR/Sustainability Committee are all statistically significant and in the expected direction. Furthermore, the moderating role of family ownership on the gender diversity effect is also confirmed. However, the moderating effect of family ownership on the board independence-CSR/Sustainability Committee relationship is not supported.

Table 5. Summary of Hypothesis Tests

Hypothesis	Expected direction	Key coefficient	Result
H1: Firms with greater board gender diversity are more likely to have a CSR/Sustainability Committee.	Positive	$b = 0.100^{***}$ (M1)	Supported
H2: Firms with greater board independence are more likely to have a CSR/Sustainability Committee.	Positive	$b = 0.042^{***}$ (M1)	Supported
H3: Family firms are less likely to have a CSR/Sustainability Committee than non-family firms	Negative	$b = -2.654^{***}$ (M1)	Supported
H4: Family firm ownership weakens the positive effect of board gender diversity on the likelihood of having a CSR/Sustainability Committee.	Negative interaction	$b = -0.080^{***}$ (M2)	Supported

Hypothesis	Expected direction	Key coefficient	Result
H5: Family firm ownership weakens the positive effect of board independence on the likelihood of having a CSR/Sustainability Committee.	Negative interaction	$b = 0.014$ n.s. (M3)	Not supported

5. Discussion

This study examined the governance determinants of CSR/Sustainability Committee adoption among 845 US publicly listed firms over the period 2018 to 2022. Building on agency theory (Jensen & Meckling, 1976) and socioemotional wealth theory (Gómez-Mejía et al., 2007), five hypotheses were tested using random-effects logit models with a Mundlak-type test of the random-effects assumption (Mundlak, 1978). Of the five hypotheses proposed, four are supported by empirical evidence. The direct effects of board gender diversity, board independence, and family firm ownership on CSR/Sustainability Committee adoption are all significant and in the expected direction. The moderating effect of family ownership on the gender diversity-committee relationship is also confirmed. However, the moderating effect of family ownership on the board independence-committee relationship is not supported.

Hypothesis 1, which predicted a positive relationship between board gender diversity and the existence of a CSR/Sustainability Committee, is supported. This result aligns with the majority of research: Studies show that female directors bring broader stakeholder-oriented values and greater sensitivity to social risks (Boulouta, 2013; Galbreath, 2011). From an agency theory perspective, gender-diverse boards also achieve stronger monitoring capacity (Adams & Ferreira, 2009), and a CSR/Sustainability Committee can be understood as a direct governance consequence of that capacity. Furthermore, the result is consistent with Martinez-Ferrero et al. (2024), who find the same positive effect of board gender diversity on the existence of a CSR/Sustainability Committee in an international sample, and with the broader literature linking female board representation to stronger CSR outcomes (Bear et al., 2010; Deschênes et al., 2015; Liao et al., 2015). This finding shows that board gender diversity is an important driver of the adoption of a CSR/Sustainability Committee.

Hypothesis 2, which predicted a positive relationship between board independence and the existence of a CSR/Sustainability Committee, is also supported. Under agency theory, independent directors are free from managerial ties (Fama & Jensen, 1983) and could therefore

be more likely to support non-financial governance structures such as a CSR/Sustainability Committee. The result is consistent with Agarwala et al. (2023), who report a positive effect of board independence on CSR performance, and with the qualitative evidence in Islam et al. (2023), where independent directors contribute to CSR decision-making through their outsider perspective and monitoring role. Martinez-Ferrero et al. (2024), by contrast, examine board independence only as a moderator that weakens the effect of board gender diversity, not as a direct predictor. This study therefore adds evidence on a direct effect of board independence on the adoption of a CSR/Sustainability Committee. This gap, to the best of our knowledge, has not been tested in the international literature. In the sample context (US companies), this finding could be justified by regulatory reforms after 2002. These reforms raised board independence requirements in the US (Chhaochharia & Grinstein, 2007), which may help explain why board independence emerges as a direct predictor of formal CSR governance in this setting.

Hypothesis 3 predicted that family-controlled firms are less likely to have a CSR/Sustainability Committee and received strong empirical support. This is consistent with socioemotional wealth theory (see Section 2.2.3): A CSR/Sustainability Committee directly threatens the non-financial priorities family owners seek to protect, which explains why family firms are less likely to adopt one (Gómez-Mejía et al., 2007; Hsueh et al., 2023; Oh et al., 2019). Furthermore, Cruz et al. (2014) provide reasons in a similar direction: Family firms do not reject CSR as such. Instead, they resist formalizing it through board-level structures that reduce their control. In sum, the finding suggests that what family firms oppose is not CSR itself, but the institutional oversight that comes with it.

Hypothesis 4, which predicted that family ownership weakens the positive effect of board gender diversity on the existence of a CSR/Sustainability Committee, is supported in both the isolated and full model. The interaction plot (Figure 2) illustrates this clearly: In non-family firms, higher board gender diversity is strongly associated with a greater likelihood of CSR/Sustainability Committee adoption. In contrast, in family-controlled firms, the relationship is nearly flat. This finding shows where agency theory could reach its limits: Female directors are generally expected to drive formal CSR governance, but their ability to do so depends on the context: In family-controlled firms, the family's interest in preserving control overrides conventional board-level governance (Cruz et al., 2014). This leaves female directors with little room to influence structural decisions such as CSR/Sustainability Committee adoption. Villalonga and Amit (2006) likewise find that the share of independent directors has

no significant effect on firm value, consistent with the notion that board members in family firms often serve a symbolic rather than a substantive function (Oh et al., 2019).

Hypothesis 5 predicted that family ownership weakens the positive effect of board independence on the existence of a CSR/Sustainability Committee. However, this hypothesis is not supported. The interaction term is not significant in either the isolated model or the full model. One possible explanation is that board independence, by regulatory and normative definition, excludes directors with direct family, employment, or economic ties to the controlling family (Anderson & Reeb, 2004). This leaves little room for family ownership to shape the preferences of independent directors, such as the adoption of a CSR/Sustainability Committee. The usual channels of family influence, namely family ties and economic dependence, are structurally absent. Board gender diversity, by contrast, is a demographic rather than an independence-based category, meaning that female representation can be increased "internally," through the appointment of women who are themselves members of the controlling family. When this is the case, their preferences may be more closely aligned with the preservation of socioemotional wealth (SEW) and family control (Gómez-Mejía et al., 2007) than with the pursuit of external legitimacy through CSR structures, which could explain why family ownership decreases the effect of gender diversity but not that of board independence. An alternative explanation is offered by Oh et al. (2019), who show that in family-managed firms boards often serve as impression management tools rather than as substantive monitoring bodies, leaving little additional influence for family ownership to constrain.

Taken together, these results suggest that family ownership does not affect all governance mechanisms uniformly: It reduces the influence of board gender diversity, while leaving the influence of board independence on CSR/Sustainability Committee adoption unaffected.

Regarding the control variables, firm size is significantly and positively associated with CSR/Sustainability Committee adoption. This likely reflects that larger firms operate under closer public scrutiny (Cowen et al., 1987) and tend to have greater access to resources to build out formal CSR governance structures (Udayasankar, 2008). Firm age also shows a significant positive association, suggesting that more established firms are more likely to have a CSR/Sustainability Committee in place. Leverage, however, is not significantly associated with a CSR/Sustainability Committee adoption in this sample.

6. Conclusion

6.1 Main Findings

In summary, this study examines which governance characteristics make firms more or less likely to have a CSR/Sustainability Committee. Corporate governance research has long asked how board characteristics shape CSR outcomes. This paper takes a different angle: Instead of looking at CSR performance, it examines the structural decision to adopt a CSR/Sustainability Committee in the first place. Moreover, it asks whether that decision depends on who owns the firm. The results show that it does: Board gender diversity (BGD) and board independence (BI) matter, but their influence is not the same in every firm. Ownership context, and in particular whether the firm is family-controlled, shapes how much influence board characteristics actually have.

6.2 Theoretical Implications

This study contributes to the theoretical literature in several ways. First, it adds to the limited body of research on the antecedents of CSR/Sustainability Committee adoption, providing evidence from a US context using recent panel data. While prior work has largely focused on the consequences of having a CSR/Sustainability Committee (e.g., Baraibar-Diez & Odriozola, 2019; Elmaghrabi, 2021; Radu & Smaili, 2022), less is known about which governance characteristics lead firms to establish one (Martinez-Ferrero et al., 2024). The findings show that board gender diversity, board independence, and family ownership are all significant predictors. This gives a clearer picture of the conditions under which formal CSR governance appears at the board level.

Second, the study extends work on how family ownership shapes formal CSR governance (Cruz et al., 2014; Hsueh et al., 2023; Oh et al., 2019). Family firms are less likely to adopt a CSR/Sustainability Committee, and family ownership also weakens the positive effect of board gender diversity. Governance characteristics therefore need to be examined in the context of the firm's ownership structure, in line with prior evidence that governance mechanisms do not affect CSR outcomes uniformly across ownership contexts (Bansal et al., 2018).

Third, the asymmetric moderating role of family ownership is theoretically important: It weakens the effect of board gender diversity but does not moderate the effect of board independence. This suggests that the impact of corporate governance on CSR/Sustainability

Committee adoption is not uniform, but conditional on which mechanism is considered and in what ownership context it operates.

6.3 Managerial Implications

The findings offer implications for several groups. For investors who value CSR and ESG practices, the results suggest that board gender diversity and board independence are relevant indicators of whether a firm has formal CSR governance in place. However, ownership structure matters. In family firms, board gender diversity is a less reliable indicator of CSR/Sustainability Committee adoption. Investors should therefore consider both board composition and ownership structure when assessing a firm's CSR governance.

For boards and nominating committees, the findings highlight that appointing female directors is not sufficient on its own. In family firms, the positive influence that gender diversity would otherwise have on CSR governance is significantly weakened. As a consequence, supervisory boards should actively ensure that female directors have real decision-making influence, not just formal representation.

Finally, the results are relevant for CSR/ESG rating agencies. Family firms are less likely to have a CSR/Sustainability Committee. However, this does not necessarily mean they are less committed to CSR. Rating agencies should therefore not only look at whether a CSR/Sustainability Committee exists, but also take ownership structure into account when assessing a firm's CSR governance.

6.4 Policy Implications

The findings also offer implications for regulators. For rules on board independence, the results suggest that it matters how independent directors are selected, not only how many there are. Anderson and Reeb (2004) show that families limit independent representation through the nominating committee. As a result, a firm can meet a minimum quota while the family still shapes who joins the board. Regulators should therefore address the independence of the nomination process itself, not only the share of independent directors.

Regarding the firms' development over time, the results show that market pressure raises adoption but does not close the gap. Compared to 2019, the likelihood of having a CSR/Sustainability Committee increases in every following year (see Appendix G). However,

the negative effect of family ownership is estimated alongside these year effects and remains strong in all models. The general trend therefore does not explain why family firms adopt fewer CSR/Sustainability Committees. Regulators should therefore not rely on market pressure alone to reach family-controlled firms.

Finally, the results are relevant for how firm size is treated in governance regulation. Larger firms are much more likely to have a CSR/Sustainability Committee, so formal CSR governance is less common among smaller firms. Regulation already accounts for this dimension: In several jurisdictions, requirements to establish board committees depend on company size, and proportionality is an established principle in their design (Rey, 2022). The results suggest that such proportionality should not translate into exempting smaller firms altogether, but into requirements that scale with firm size.

6.5 Limitations

Several limitations of this study should be acknowledged. First, the dependent variable is binary and only captures whether a CSR/Sustainability Committee exists. It does not measure the committee's actual influence or effectiveness. Future research could therefore examine whether the governance characteristics identified here also determine how much impact a CSR/Sustainability Committee has in practice.

Second, the dataset is limited to US publicly listed firms over the period 2018 to 2022. Therefore, the findings cannot be generalized to other countries or regions. Moreover, the observation period includes the COVID-19 pandemic. Prior research shows that ESG-rated stocks behaved atypically during the COVID-19 market crash (Albuquerque et al., 2020), which may have influenced firms' sustainability-related decisions during the sample period as well.

Finally, the study treats family firms as a homogeneous group. However, founder-led firms and later-generation family firms may behave differently. As prior research suggests, first- and later-generation family firms differ in how strongly external actors shape firm behavior (Cruz & Nordqvist, 2012). Therefore, future research could examine whether the negative effect of family ownership on CSR/Sustainability Committee adoption weakens across generations.

6.6 Future Research Directions

The findings also point to several directions for future research. First, future research could extend this study to other national contexts. The findings are based on US firms, where governance norms and regulatory requirements are well-established. However, it remains unclear whether the same patterns hold in institutional environments with weaker independence standards or different ownership traditions.

Furthermore, future research could compare the CSR/ESG performance of family firms with and without a CSR/Sustainability Committee. Given that family firms are less likely to adopt such structures, it remains an open question whether this comes at a cost in terms of sustainability outcomes.

Finally, this study uses a family firm dummy to proxy for socioemotional wealth and the resistance to formal governance structures that follows from it. Future research could instead use the FIBER scale (Berrone et al., 2012), which measures five specific dimensions of SEW: (F) Family Control, (I) Identification with the Firm, (B) Binding Social Ties, (E) Emotional Attachment, and (R) Renewal through Dynastic Succession. This would allow researchers to examine which dimension most strongly drives that resistance.

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Appendices

Appendix A. Sample Selection Procedure

Step	Firms	Firm-years	Change
Initial panel from Refinitiv Eikon (2018-2022, balanced)	998	4,990	-
Less: firms with fewer than three usable observations after the one-year lag	876	4,380	-610
Less: listwise deletion on remaining explanatory and control variables	846	2,841	-1,539
Less: industry 18 (Educational Services), which predicts the outcome perfectly	845	2,837	-4

Note: The one-year lag restricts the dependent variable to 2019–2022, so a firm can contribute at most four observations. In the final estimation sample, observations per firm range from one to four (mean = 3.4), as listwise deletion is applied after the three-observation filter.

Appendix B. Industry Distribution (NAICS Sectors)

Code	NAICS sector	Firms	Percent
2	Manufacturing	334	33.8
3	Finance and Insurance	143	14.5
1	Retail Trade	82	8.3
4	Information	75	7.6
5	Wholesale Trade	56	5.7
8	Professional, Scientific, and Technical Services	50	5.1
7	Transportation and Warehousing	41	4.1
12	Utilities	41	4.1
9	Mining, Quarrying, and Oil and Gas Extraction	32	3.2
10	Real Estate and Rental and Leasing	32	3.2
14	Construction	29	2.9
13	Accommodation and Food Services	25	2.5
6	Health Care and Social Assistance	21	2.1
15	Administrative and Support and Waste Management Services	19	1.9

Code	NAICS sector	Firms	Percent
16	Other Services (except Public Administration)	4	0.4
17	Arts, Entertainment, and Recreation	3	0.3
11	Agriculture, Forestry, Fishing and Hunting	1	0.1
18	Educational Services	1	0.1
Total		989	100.0

Note: 9 of the 998 firms in the initial panel carry no sector code.

Appendix C. Extended Descriptive Statistics

Variable	N	Mean	SD	Min	p25	Median	p75	Max
CSR Committee (0/1)	3,901	0.608	0.488	0.000	0.000	1.000	1.000	1.000
Board Gender Diversity (%)	3,904	25.328	10.467	0.000	18.182	25.000	30.769	75.000
Board Independence (%)	3,904	81.876	12.055	0.000	77.778	85.714	90.909	100.000
Family Firm (0/1)	4,302	0.189	0.392	0.000	0.000	0.000	0.000	1.000
Firm Size (ln Total Assets)	4,317	23.157	1.470	10.127	22.111	22.986	24.013	28.951
Leverage (Net Debt/EBITDA)	4,198	3.943	34.751	-108.568	0.589	1.905	3.683	1773.557
Firm Age (years)	4,380	67.117	47.403	0.000	29.000	52.000	102.000	230.000

Note: Firm-year observations, unlagged, before construction of the estimation sample. Observation counts differ across variables because of missing values. The extreme leverage range reflects firms with near-zero or negative EBITDA.

Appendix D. Panel Variation: Within- and Between-Firm Decomposition

Variable	Mean	Overall SD	Between SD	Within SD	N	n (firms)
CSR Committee	0.608	0.488	0.408	0.275	3,901	876
Board Gender Diversity, t-1	24.621	10.386	9.077	5.066	3,422	876
Board Independence, t-1	81.684	12.189	11.419	4.258	3,422	876
Family Firm, t-1	0.189	0.392	0.393	0.035	3,441	863
Firm Size (ln Total Assets), t-1	23.105	1.485	1.456	0.301	3,456	868
Leverage (Net Debt/EBITDA), t-1	4.283	38.556	23.790	32.012	3,357	859
Firm Age (years), t-1	66.601	47.405	47.412	1.115	3,504	876

Note: Output of the Stata command xtsum, computed on the intermediate panel of 876 firms. For the family firm indicator, almost all variation lies between firms ($SD = 0.393$) rather than within them over time ($SD = 0.035$), so a fixed-effects estimator would discard most of it.

Appendix E. Mundlak-Augmented RE Logit and Joint Significance Test

Variable	Coefficient	Std. Error
Board Gender Diversity, t-1	0.044**	(0.022)
Board Independence, t-1	-0.014	(0.022)
Family Firm, t-1	2.754	(2.281)
Firm Size (ln Total Assets), t-1	1.991***	(0.508)
Leverage (Net Debt/EBITDA), t-1	0.000	(0.002)
Firm Age (years), t-1	2.957*	(1.641)
Mean: Gender Diversity	0.121***	(0.028)
Mean: Board Independence	0.094***	(0.025)
Mean: Family Firm	-4.828**	(2.313)
Mean: Firm Size	0.438	(0.512)
Mean: Leverage	-0.017**	(0.008)
Mean: Firm Age	-2.951*	(1.641)
Constant	-62.457***	(4.072)
Observations	2,837	
Number of firms	845	
Wald χ^2 (df)	746.14 (30)	
Log likelihood	-913.734	
rho	0.919	
Joint test of firm means: $\chi^2(6)$	50.64***	

Note: Year and industry fixed effects are included but suppressed. The six within-firm means are jointly significant ($\chi^2(6) = 50.64$, $p < 0.001$), indicating that unobserved firm heterogeneity is correlated with the explanatory variables. The random-effects specification is nevertheless retained; see Section 3.4 for the reasoning. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix F. Odds Ratios for the Main Models

Variable	M1	M2	M3	M4
	Direct	H4	H5	Full
Board Gender Diversity, t-1	1.106*** (0.014)			
Board Independence, t-1	1.043*** (0.011)			
Gender Diversity (centered), t-1		1.125*** (0.017)	1.105*** (0.014)	1.127*** (0.020)
Board Independence (centered), t-1		1.045*** (0.011)	1.039*** (0.013)	1.038** (0.015)
Family Firm, t-1	0.070*** (0.024)	0.056*** (0.019)	0.076*** (0.028)	0.058*** (0.025)
Family x Gender Diversity, t-1		0.923*** (0.026)		0.912*** (0.030)
Family x Board Independence, t-1			1.014 (0.024)	1.033 (0.028)
Firm Size (ln Total Assets), t-1	12.601*** (1.574)	12.941*** (1.617)	12.662*** (1.582)	13.280*** (1.967)
Leverage (Net Debt/EBITDA), t-1	0.999 (0.002)	1.000 (0.002)	0.999 (0.002)	1.000 (0.002)
Firm Age (years), t-1	1.009*** (0.003)	1.008*** (0.003)	1.008*** (0.003)	1.009** (0.004)
Observations	2,837	2,837	2,837	2,837
Number of firms	845	845	845	845

Note: Exponentiated coefficients from the models in Appendix G; standard errors in parentheses. Values above one show higher odds of having a CSR/Sustainability Committee. Year and industry fixed effects are included but suppressed. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix G. Full RE Logit Results Including Year and Industry Fixed Effects

Variable	M1	M2	M3	M4
	Direct	H4	H5	Full
Board Gender Diversity, t-1	0.100*** (0.013)			
Board Independence, t-1	0.042*** (0.011)			
Family Firm, t-1	-2.654*** (0.343)	-2.887*** (0.345)	-2.580*** (0.365)	-2.853*** (0.426)
Firm Size (ln Total Assets), t-1	2.534*** (0.125)	2.560*** (0.125)	2.539*** (0.125)	2.586*** (0.148)
Leverage (Net Debt/EBITDA), t-1	-0.001 (0.002)	-0.000 (0.002)	-0.001 (0.002)	-0.000 (0.002)
Firm Age (years), t-1	0.009*** (0.003)	0.008*** (0.003)	0.008*** (0.003)	0.009** (0.004)
Gender Diversity (centered), t-1		0.118*** (0.015)	0.100*** (0.013)	0.120*** (0.017)
Board Independence (centered), t-1		0.044*** (0.011)	0.039*** (0.013)	0.037** (0.015)
Family x Gender Diversity, t-1		-0.080*** (0.028)		-0.092*** (0.033)
Family x Board Independence, t-1			0.014	0.033

Variable	M1	M2	M3	M4
	Direct	H4	H5	Full
			(0.023)	(0.028)
Year 2020	1.445*** (0.225)	1.427*** (0.225)	1.447*** (0.225)	1.584*** (0.243)
Year 2021	3.143*** (0.264)	3.102*** (0.264)	3.149*** (0.264)	3.411*** (0.300)
Year 2022	4.619*** (0.343)	4.557*** (0.342)	4.631*** (0.342)	4.921*** (0.394)
Industry 2 (Manufacturing)	3.223*** (0.516)	3.313*** (0.516)	3.222*** (0.516)	3.442*** (0.602)
Industry 3 (Finance and Insurance)	-3.190*** (0.609)	-3.206*** (0.608)	-3.192*** (0.609)	-3.147*** (0.713)
Industry 4 (Information)	-2.751*** (0.677)	-2.856*** (0.677)	-2.736*** (0.676)	-2.832*** (0.792)
Industry 5 (Wholesale Trade)	-0.154 (0.702)	-0.085 (0.702)	-0.173 (0.702)	-0.160 (0.820)
Industry 6 (Health Care and Social Assistance)	-3.437*** (0.873)	-3.462*** (0.871)	-3.452*** (0.872)	-3.584*** (1.022)
Industry 7 (Transportation and Warehousing)	2.201*** (0.725)	2.227*** (0.725)	2.194*** (0.724)	2.338*** (0.852)
Industry 8 (Professional, Scientific, and Technical Services)	2.778*** (0.744)	2.818*** (0.744)	2.771*** (0.743)	2.895*** (0.870)
Industry 9 (Mining, Quarrying, and Oil and Gas Extraction)	6.862***	6.990***	6.850***	6.933***

Variable	M1	M2	M3	M4
	Direct	H4	H5	Full
	(1.137)	(1.143)	(1.137)	(1.259)
Industry 10 (Real Estate and Rental and Leasing)	5.281***	5.336***	5.296***	5.391***
	(1.029)	(1.028)	(1.025)	(1.144)
Industry 12 (Utilities)	5.042***	5.073***	5.063***	5.082***
	(1.071)	(1.072)	(1.070)	(1.212)
Industry 13 (Accommodation and Food Services)	1.231	1.232	1.220	1.240
	(0.859)	(0.858)	(0.858)	(0.997)
Industry 14 (Construction)	-1.089	-1.198	-1.100	-1.325
	(0.807)	(0.805)	(0.806)	(0.955)
Industry 15 (Administrative and Support and Waste Management Services)	2.056**	2.055**	2.064**	2.141*
	(0.940)	(0.939)	(0.940)	(1.109)
Industry 16 (Other Services (except Public Administration))	4.778*	4.738*	4.797*	4.580
	(2.837)	(2.871)	(2.836)	(4.280)
Industry 17 (Arts, Entertainment, and Recreation)	-7.864**	-7.888**	-7.916**	-8.303**
	(3.282)	(3.309)	(3.277)	(3.874)
Constant	-64.039***	-58.684***	-58.216***	-59.358***
	(2.956)	(2.872)	(2.870)	(3.415)
Observations	2,837	2,837	2,837	2,837
Number of firms	845	845	845	845
Wald χ^2 (df)	890.74 (24)	906.58 (25)	897.67 (25)	664.26 (26)
Log likelihood	-916.615	-915.759	-916.178	-925.899
sigma_u	6.055	6.101	6.062	6.480

Variable	M1	M2	M3	M4
	Direct	H4	H5	Full
rho	0.918	0.919	0.918	0.927
LR test of rho = 0 (chibar2)	943.92	944.43	944.68	924.14

Note: Log-odds coefficients from random-effects logistic regression (xtlogit, re, intpoints(7)); standard errors in parentheses. All explanatory variables are lagged by one year; M2–M4 use mean-centered governance variables in the interaction terms. Reference categories: industry 1 (Retail Trade) and year 2019. Industry 18 (Educational Services) predicts the outcome perfectly and is dropped by the estimator; the single firm in industry 11 (Agriculture) does not survive the sample filters and is therefore absent from the estimation sample. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix H. Declaration on the Use of Artificial Intelligence

While writing this thesis I used AI as a support tool. It helped me to correct the language and to improve the clarity and academic style of the text I had written myself. I used it to check my Stata analysis to include all necessary steps. Moreover, it helped me to build the appendix tables from my Stata output, and it went through the final document for spelling, grammar and APA formatting.

Everything that makes up the substance of this thesis is my own work. I personally carried out and critically reviewed all aspects of this thesis, including research design, data collection, statistical analysis, interpretation of results, and final formulation. AI tools were at no point used to independently conduct the statistical analysis or generate content.

Examples of prompts used in Claude by Anthropic (translated from German):

- "Please check the entire thesis for spelling and grammar (US English)."
- "Please check this thesis on APA Guidelines and provide possible improvements"
- "Please check whether this paragraph is written in appropriate academic English and suggest improvements."
- "Go through this Stata output code and check whether the code includes all the necessary steps for this binary regression analysis."
- "Build this table from this Stata output (e.g. Descriptive Statistic Table) as a Word file."